Chapter 11. Variable Selection

Consequences of omitting variables, search methods, selection criteria, Mallows' C_p

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With many candidate regressors we want a **parsimonious** model: enough variables to explain and predict y, no more, since each adds variance.

11.1 Consequences of omitting variables

Partition the **full model** as

$$\mathbf{y} = X\beta + \epsilon = X_k\beta_k + X_r\beta_r + \epsilon, \qquad \epsilon \sim N(\mathbf{0}, \sigma^2 I),$$

where X_k $(n \times (k+1))$ contains the intercept and k regressors and X_r $(n \times r,\ r = p-k)$ the others. The full-model estimator $\hat\beta = (\hat\beta_k', \hat\beta_r')'$ and $\hat\sigma^2 = SSE/(n-p-1)$ are compared with the **reduced model** estimators

$$\tilde\beta_k = (X_k'X_k)^{-1}X_k'\mathbf{y}, \qquad \tilde\sigma^2 = \frac{\mathbf{y}'\{I - X_k(X_k'X_k)^{-1}X_k'\}\mathbf{y}}{n-k-1}.$$

Theorem 11.1

Let $A = (X_k'X_k)^{-1}X_k'X_r$ (the **alias matrix**) and $H_k = X_k(X_k'X_k)^{-1}X_k'$. If the full model is true:

(a) $E\tilde\beta_k = \beta_k + A\beta_r$ and $\operatorname{Var}\tilde\beta_k = \sigma^2(X_k'X_k)^{-1}$; so $\operatorname{MSE}(\tilde\beta_k) = E(\tilde\beta_k - \beta_k)(\tilde\beta_k - \beta_k)' = \sigma^2(X_k'X_k)^{-1} + A\beta_r\beta_r'A'$.

(b) $E\tilde\sigma^2 = \sigma^2 + \dfrac{\beta_r'X_r'(I-H_k)X_r\beta_r}{n-k-1} \geq \sigma^2$.

(c) For predicting $E(y) = \mathbf{x}'\beta = \mathbf{x}_k'\beta_k + \mathbf{x}_r'\beta_r$ by $\tilde y = \mathbf{x}_k'\tilde\beta_k$:

$$\operatorname{MSE}(\tilde y) = \sigma^2\mathbf{x}_k'(X_k'X_k)^{-1}\mathbf{x}_k + (\mathbf{x}_k'A\beta_r - \mathbf{x}_r'\beta_r)^2.$$

Proof. (a) $E\tilde\beta_k = (X_k'X_k)^{-1}X_k'(X_k\beta_k + X_r\beta_r) = \beta_k + A\beta_r$; the variance is as usual; MSE = Var + bias · bias′.

(b) Theorem 5.4(a) with $I - H_k$, whose trace is $n-k-1$ and which annihilates $X_k\beta_k$.

(c) MSE = Var + bias^2 with bias $\mathbf{x}_k'(\beta_k + A\beta_r) - \mathbf{x}'\beta$. ■

So omitting relevant variables biases the remaining coefficients (unless $X_k'X_r = 0$) and inflates the residual variance estimate. In exchange the variance decreases:

Theorem 11.2

(a) $\operatorname{Var}(\hat\beta_k) - \operatorname{Var}(\tilde\beta_k)$ is nonnegative definite.

(b) $\operatorname{Var}(\mathbf{x}'\hat\beta) \geq \operatorname{Var}(\mathbf{x}_k'\tilde\beta_k)$.

(c) If $\operatorname{Var}(\hat\beta_r) - \beta_r\beta_r'$ is nonnegative definite, then $\operatorname{Var}(\hat\beta_k) - \operatorname{MSE}(\tilde\beta_k)$ is nonnegative definite and $\operatorname{Var}(\mathbf{x}'\hat\beta) \geq \operatorname{MSE}(\mathbf{x}_k'\tilde\beta_k)$.

Proof. By the partitioned-inverse formula, with $B = X_r'(I-H_k)X_r$ (nonsingular),

$$\sigma^{-2}\operatorname{Var}(\hat\beta) = (X'X)^{-1} = \begin{pmatrix} (X_k'X_k)^{-1} + AB^{-1}A' & -AB^{-1} \\ -B^{-1}A' & B^{-1} \end{pmatrix}.$$

(a) $\operatorname{Var}\hat\beta_k - \operatorname{Var}\tilde\beta_k = \sigma^2 AB^{-1}A' \geq 0$.

(b) With $\mathbf{x} = (\mathbf{x}_k', \mathbf{x}_r')'$ and $\mathbf{u} = A'\mathbf{x}_k - \mathbf{x}_r$,

$$\sigma^{-2}\operatorname{Var}(\mathbf{x}'\hat\beta) = \mathbf{x}_k'(X_k'X_k)^{-1}\mathbf{x}_k + \mathbf{u}'B^{-1}\mathbf{u} \geq \mathbf{x}_k'(X_k'X_k)^{-1}\mathbf{x}_k.$$

(c) $\operatorname{Var}\hat{\beta}_r = \sigma^2 B^{-1}$, so the assumption reads $\sigma^2 B^{-1} \geq \beta_r \beta_r'$. Then $\operatorname{Var}\hat{\beta}_k - \operatorname{MSE}(\tilde{\beta}_k) = A(\sigma^2 B^{-1} - \beta_r \beta_r')A' \geq 0$, and similarly $\operatorname{Var}(\mathbf{x}'\hat{\beta}) - \operatorname{MSE}(\mathbf{x}_k'\tilde{\beta}_k) = \mathbf{u}'(\sigma^2 B^{-1} - \beta_r \beta_r')\mathbf{u} \geq 0$. ■

So dropping variables with coefficients small relative to their SEs ($\beta_r \beta_r' \leq \operatorname{Var}\hat{\beta}_r$) reduces MSE despite bias; if $\beta_r = \mathbf{0}$, both are unbiased and the reduced model has smaller variance.

11.2 Search methods

With p candidate regressors there are 2^p subsets (each variable in or out), so exhaustive search becomes infeasible for large p ($2^{50} \approx 10^{15}$).

All possible regressions (best subset). Fit all 2^p models; for each size k keep the best by R_k^2 (equivalently SSE_k, MSE_k), then compare sizes with a penalized criterion (Section 11.3).

Backward elimination.

(1) Fit the full model.

(2) Compute the partial F statistic of each variable in the model.

(3) Let F_L be the smallest, and F_C the critical value at level α_{drop}.

(4) If $F_L < F_C$, delete that variable, refit and return to (2); otherwise stop.

This needs at most $p(p+1)/2$ fits.

Forward selection. Start empty; add the variable with the largest partial F if it exceeds the level-α_{add} critical value; else stop.

Stepwise selection. Forward selection, re-examining included variables after each addition and removing those below α_{drop} ($\alpha_{\text{drop}} \geq \alpha_{\text{add}}$ to avoid cycling).

The methods may disagree, none guarantees the best subset, and the levels are not true significance levels.

Example 11.1

Hald's cement data ($n = 13$, ingredients $x_1, \ldots, x_4$, $x_1 + x_2 + x_3 + x_4 \approx 100$, x_2, x_4 highly correlated): backward elimination ($\alpha = 0.05$) removes x_3 (partial $F = 0.02$), then x_4 ($1.86 < F_{0.05}(1, 9) = 5.12$), ending at $\{x_1, x_2\}$ ($R^2 = 0.979$). Forward selection enters x_4 ($R^2 = 0.675$), then x_1, and stops (x_2: $5.03 < 5.12$) at $\{x_1, x_4\}$ ($R^2 = 0.972$). Stepwise ($\alpha_{\text{add}} = 0.10$) enters x_2, removes x_4, ending at $\{x_1, x_2\}$: nearly equivalent information gives different, almost equally good models.

11.3 Selection criteria

Motivation. R^2 cannot choose a size; MSE, adjusted R^2 and Mallows' C_p penalize complexity.

For a model with k regressors (plus intercept):

(1) residual mean square $MSE_k = SSE_k/(n - k - 1)$;

(2) $R_k^2 = 1 - SSE_k/SST$;

(3) adjusted R^2: $R_{a,k}^2 = 1 - \dfrac{n-1}{n-k-1}(1-R_k^2) = 1 - \dfrac{MSE_k}{SST/(n-1)}$;

(4) Mallows' $C_k = \dfrac{SSE_k}{\hat\sigma^2} + 2(k+1) - n$, with $\hat\sigma^2$ the MSE of the full model.

R_k^2 never decreases, so it compares only equal sizes. MSE_k can increase with a variable (a lost d.f. may outweigh the drop in SSE), and maximizing $R_{a,k}^2$ is minimizing MSE_k.

Mallows' C_p. We judge the reduced model by the total standardized mean squared error of its fitted values at the n design points:

$$\Gamma_k = \frac{1}{\sigma^2}\sum_{i=1}^{n} \mathrm{MSE}(\tilde y_i), \qquad \tilde y_i = \mathbf{x}_{ki}'\tilde\beta_k.$$

Theorem 11.3

$$\Gamma_k = (k+1) + \frac{\beta_r' X_r'(I-H_k)X_r\beta_r}{\sigma^2} = \frac{E(SSE_k)}{\sigma^2} + 2(k+1) - n.$$

Hence C_k estimates Γ_k, and if the reduced model is correct ($\beta_r = \mathbf{0}$), $E(SSE_k) = (n-k-1)\sigma^2$ and $C_k \approx k+1$.

Proof [Supplement]. By Theorem 11.1(c), the variance part sums to

$$\sum_i \mathbf{x}_{ki}'(X_k'X_k)^{-1}\mathbf{x}_{ki} = \mathrm{tr}\Big\{(X_k'X_k)^{-1}\sum_i \mathbf{x}_{ki}\mathbf{x}_{ki}'\Big\} = \mathrm{tr}\{(X_k'X_k)^{-1}X_k'X_k\} = k+1.$$

The bias vector over the design points is $X_kA\beta_r - X_r\beta_r = -(I-H_k)X_r\beta_r$, with squared length $\beta_r'X_r'(I-H_k)X_r\beta_r$. By Theorem 11.1(b), this equals $E(SSE_k) - (n-k-1)\sigma^2$. Substituting gives the second expression. ■

So we look for models with small C_k and C_k close to $k+1$; a C_k well above $k+1$ signals bias from omitted variables. For the full model, $C_p = p+1$ identically.

11.4 Selected exercises

Motivation. Omitted-variable bias for a line, and adjusted R^2 and C_k via the partial F.

Exercise 11.1 (Omitted-variable bias for a straight line)

The straight line $\hat y = \hat\beta_0 + \hat\beta_1 x_1$ is fitted by least squares, but the true mean is $E(y) = \beta_0 + \beta_1 x_1 + \beta_2 x_2$ (errors uncorrelated with variance σ^2). Write $S_{11} = \sum(x_{i1}-\bar x_1)^2 > 0$, $S_{22} = \sum(x_{i2}-\bar x_2)^2$ and $S_{12} = \sum(x_{i1}-\bar x_1)(x_{i2}-\bar x_2)$.
(a) Is $\hat\beta_1$ biased? Find $E\hat\beta_1$.
(b) Find the expectation of the residual mean square of the fitted line.

Solution.

(a) With $k_i = (x_{i1}-\bar x_1)/S_{11}$, $\hat\beta_1 = \sum k_i y_i$, $\sum k_i = 0$ and $\sum k_i x_{i1} = 1$ (Section 4.1). Hence

$$E\hat\beta_1 = \sum k_i(\beta_0 + \beta_1 x_{i1} + \beta_2 x_{i2}) = \beta_1 + \beta_2\sum k_i x_{i2} = \beta_1 + \beta_2\frac{S_{12}}{S_{11}}.$$

So $\hat{\beta}_1$ is biased unless $\beta_2 = 0$ or $S_{12} = 0$. S_{12}/S_{11}, the slope of x_2 on x_1, is the alias-matrix entry of Theorem 11.1: $\hat{\beta}_1$ absorbs the part of x_2's effect travelling with x_1.

(b) By Theorem 11.1(b) with $k = 1$ and $X_r = \mathbf{x}_2$, the excess is $\beta_2^2\, \mathbf{x}_2'(I - H_k)\mathbf{x}_2/(n-2)$, where $\mathbf{x}_2'(I - H_k)\mathbf{x}_2$ is the residual sum of squares of the regression of x_2 on $(1, x_1)$, namely $S_{22} - S_{12}^2/S_{11}$ (Theorem 3.3). So

$$E(MSE) = \sigma^2 + \frac{\beta_2^2(S_{22} - S_{12}^2/S_{11})}{n-2} \geq \sigma^2.$$

■

Exercise 11.2 (Adjusted R^2 and the partial F statistic)

A model with k regressors (and intercept) is enlarged by r more; F is their partial F with the larger model's residual mean square. With $SSE_{k+r} > 0$, show that adjusted R^2 increases iff $F > 1$.

Solution [Supplement].

Step 1. Since $R_a^2 = 1 - MSE/\{SST/(n-1)\}$ with SST common to both models, R_a^2 increases iff $MSE_{k+r} < MSE_k$.

Step 2. By definition $F = \{(SSE_k - SSE_{k+r})/r\}/MSE_{k+r}$, so $SSE_k = SSE_{k+r} + rF\, MSE_{k+r} = (n - k - r - 1 + rF)MSE_{k+r}$, and

$$MSE_k = \frac{SSE_k}{n-k-1} = \frac{n-k-r-1+rF}{n-k-1}\, MSE_{k+r}.$$

Step 3. With $MSE_{k+r} > 0$, $MSE_k > MSE_{k+r}$ iff $n - k - r - 1 + rF > n - k - 1$ iff $F > 1$. So adjusted R^2 admits a variable far more easily than a significance test, which would require $F > F_\alpha$ (about 4 for one variable at $\alpha = 0.05$). ■

Exercise 11.3 (Mallows' C_k and the partial F statistic)

Let F_0 be the partial F statistic for dropping the $r = p - k$ regressors of X_r from the full model. Show that

$$C_k = (k+1) + (p-k)(F_0 - 1),$$

so that $C_k \leq k + 1$ if and only if $F_0 \leq 1$.

Solution [Supplement]. Let $\hat{\sigma}^2 = SSE/(n-p-1)$ be the full-model residual mean square. By definition of F_0, $SSE_k = SSE + (p-k)F_0\hat{\sigma}^2 = \{(n-p-1) + (p-k)F_0\}\hat{\sigma}^2$. Substituting into $C_k = SSE_k/\hat{\sigma}^2 + 2(k+1) - n$,

$$C_k = (n-p-1) + (p-k)F_0 + 2(k+1) - n = (k+1) + (p-k)F_0 - (p-k).$$

Since $F_0 \approx 1$ on average when $\beta_r = \mathbf{0}$ (its mean is $\nu/(\nu-2)$ with $\nu = n-p-1$), this matches $C_k \approx k + 1$ in Theorem 11.3. ■